

Code and data

Code and data - canonicalmg, GitHub.

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Content

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README.md

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Reverse CAPTCHA: Evaluating LLM Susceptibility to Invisible Unicode Instruction Injection

An evaluation framework that tests whether large language models follow invisible Unicode-encoded instructions embedded in otherwise normal-looking text. Where traditional CAPTCHAs exploit tasks humans can solve but machines cannot, Reverse CAPTCHA exploits a perception channel that machines can access but humans cannot.

Paper: [paper/reverse_captcha.pdf](#)

Key Findings

Across **8,308 graded outputs** from 5 models (GPT-5.2, GPT-4o-mini, Claude Opus 4, Claude Sonnet 4, Claude Haiku 4.5):

Tool use is the primary amplifier

Code execution access transforms invisible Unicode from an ignorable artifact to a decodable instruction channel. Without tools, compliance stays below 17%. With tools and hints, it reaches 98-100%.

[\[image: Tool Use Ablation\]](#)

Encoding vulnerability is provider-specific

OpenAI models decode zero-width binary but not Unicode Tags. Anthropic models prefer Unicode Tags. An attacker must tailor encoding to the target provider.

[\[image: Encoding Comparison\]](#)

Full compliance heatmap

The hint gradient is consistent across models: unhinted << codepoint hints < full hints. The combination of tool access and decoding instructions is the critical enabler.

[\[image: Compliance Heatmap\]](#)

How It Works

Two invisible encoding schemes are embedded in trivia questions:

- **Zero-Width Binary (ZW):** Each ASCII character encoded as 8 binary digits using U+200B (0) and U+200C (1)
- **Unicode Tags:** Each ASCII character mapped to U+E0000 + codepoint (one invisible char per ASCII char)

The model receives a trivia question with a known visible answer. Hidden in the invisible characters is a different answer. If the model outputs the hidden answer, it followed the invisible instruction.

Test Matrix

- 30 trivia Q/A pairs
- 2 encoding schemes x 4 hint levels + 1 control = 9 conditions
- 2 tool conditions (ON/OFF)
- 5 models
- 270 cases per eval run

Statistical Analysis

- Wilson score 95% confidence intervals
- Fisher's exact test with Bonferroni correction
- Chi-squared tests for per-model scheme variation
- Cohen's h effect sizes

Quick Start

```
# Install
pip install -e ".[dev]"

# Run reverse CAPTCHA eval
evalrun run --pack reverse_captcha --model openai:gpt-4o-mini --tools --n 3 --out
results.sqlite

# Run without tools
evalrun run --pack reverse_captcha --model openai:gpt-4o-mini --n 3 --out results.sqlite

# Run with Anthropic models
evalrun run --pack reverse_captcha --model anthropic:claude-sonnet-4-20250514 --tools --n
3 --out results.sqlite

# Generate analysis
python3 scripts/analyze_journal.py

# Generate figures
python3 scripts/generate_figures.py
```

Project Structure

└─ packs/reverse_captcha/	# Eval pack (cases, grader, generator)
│ └─ cases.yaml	# 270 test cases
│ └─ qa_pairs.yaml	# 30 trivia Q/A pairs
│ └─ grader.py	# Deterministic grading (5 categories)
│ └─ generate_cases.py	# Case generator for both encodings
└─ src/evalrun/	# Core framework
│ └─ adapters/	# OpenAI + Anthropic API adapters
│ └─ cli.py	# CLI (run, report, export)
│ └─ runner.py	# Eval runner with tool-use support
│ └─ db.py	# SQLite persistence
└─ paper/	# Workshop paper
│ └─ reverse_captcha.tex	# LaTeX source
│ └─ reverse_captcha.pdf	# Compiled PDF
└─ results/journal/	# Raw data
│ └─ analysis/	# Statistical analysis CSVs
│ └─ figures/	# Publication figures (PDF + PNG)
└─ scripts/	# Analysis and figure generation
└─ tests/	# Test suite

Other Eval Packs

The framework also includes two additional packs:

- **Watermark Robustness** — Tests whether models preserve a provenance marker across rewriting tasks
- **Hidden Message Extraction** — Tests whether models can extract hidden messages given explicit rules

Environment

- Python 3.10+
- `OPENAI_API_KEY` for OpenAI models
- `ANTHROPIC_API_KEY` for Anthropic models
- Dependencies: `click`, `openai`, `anthropic`, `pyyaml`, `matplotlib`, `pandas`, `scipy`, `numpy`

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